VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA

A PROJECT REPORT

on

"FACIAL ANALYSIS POWERED VIRTUAL HEALTHCARE"

Submitted in partial fulfilment of the requirements for the award of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE & ENGINEERING

Submitted By

Name	USN
SAMEEKSHA	4VP21CS078
SWASTHIKA RAI B S	4VP21CS105
THANMAYEE J	4VP21CS106
POOJA A	4VP22CS406

Under the Guidance of

Prof. THAPASWINI P S

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE &ENGINEERING

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt., of Karnataka

Nehru Nagar, Puttur - 574 203, DK, Karnataka, India.

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Nehru Nagar, Puttur - 574203, DK, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

Certified that the project work entitled "**Facial Analysis Powered Virtual Healthcare**" is carried out by **Miss. Sameeksha, Miss. Swasthika Rai B S, Miss. Thanmayee J, Miss. Pooja A** bearing USNs **4VP21CS078**, **4VP21CS105, 4VP21CS106 and 4VP22CS406** respectively bonafide students of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering** in **Computer Science & Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2024-25. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of the Guide **Prof. Thapaswini P S**	**Signature of the HOD** **Prof. Pradeep Kumar K G**	**Signature of the Principal** **Dr. Mahesh Prasanna K**

EXTERNAL VIVA

Name of the Examiners	Signature with date
1…….....	.....
2…….....	.....

DECLARATION

We, **Miss. Sameeksha(4VP21CS078), Miss. Swasthika Rai B S(4VP21CS105), Miss. Thanmayee J(4VP21CS106), Miss. Pooja A(4VP22CS406)** students of B.E. 8th Semester in Computer Science &Engineering, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled **"Facial Analysis Powered Virtual Healthcare"** has been carried out by us at VCET, Puttur, under the guidance of **Prof. Thapaswini P. S.,** Assistant Professor, Department of Computer Science &Engineering, Vivekananda College of Engineering & Technology, Puttur, and submitted in partial fulfilment of the requirements for the award of degree in **Bachelor of Engineering in Computer Science &Engineering** by **Visvesvaraya Technological University**, Belagavi during the academic year 2024-2025.

Name of the students	USN	Signature with date
Sameeksha	4VP21CS078	
Swasthika Rai B S	4VP21CS105	
Thanmayee J	4VP21CS106	
Pooja A	4VP22CS406	

Date:

Place: Puttur

ABSTRACT

Mental health awareness is growing globally, yet many individuals suffering from conditions like depression remain undiagnosed due to stigma, lack of access, or hesitance to seek professional help. This project presents an AI-based system designed to assist counselors in providing more effective, real-time support to their clients. The application combines facial expression recognition, voice analysis, and intelligent chatbot communication to detect emotional states and possible signs of depression. Using deep learning models, particularly Convolutional Neural Networks (CNNs), the system analyzes facial expressions captured via camera to identify key emotional cues such as sadness, anger, or anxiety. In parallel, a voice assistant evaluates vocal tone, pitch, and response patterns to detect behavioral signs commonly associated with mental distress. Based on these multimodal inputs, the system generates emotional insights which are shared with the counselor to support accurate assessment and intervention. A built-in chatbot module also allows clients to engage in secure, empathetic conversations between counseling sessions. It provides mental health tips, responds to common queries, and encourages users to communicate their feelings, ensuring ongoing emotional engagement and early detection of critical signs. This tool is specifically developed for use by mental health professionals and counselors, enabling them to offer clients a hybrid model of support that combines in-person expertise with intelligent virtual assistance. The system not only enhances the counselor's ability to monitor emotional well-being over time but also empowers clients with accessible, non-intrusive mental health support. The overall goal is to create a bridge between traditional therapy and modern technology for more personalized and proactive mental health care.

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INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO PROJECT

Depression and other mental health conditions have emerged as significant global health challenges, impacting diverse populations regardless of age. Prompt identification and immediate support play vital roles in effectively addressing these disorders. Conventional evaluation techniques primarily utilize self-assessment tools or direct clinical interviews. Commonly inefficient processes and may not reliably reflect current emotional conditions. To overcome these limitations, our initiative presents a virtual care platform utilizing artificial intelligence that employs facial recognition technology and speech analysis to deliver preliminary mental health evaluations.

The solution is specifically created for mental health practitioners, allowing them to implement a streamlined, evidence-based method for psychological assessment. During interaction with the system, it initiates immediate facial expression scanning through machine learning algorithms designed to identify various emotional responses happy, sad, fearful, angry, surprised, and disgusted. Supplementary vocal analysis examines speech characteristics, intonation, and emotional indicators. The integrated findings produce an initial evaluation suggesting possible depressive symptoms or emotional disturbances, followed by tailored advice ranging from relaxation techniques to recommendations for professional support.

The emotion and depression-focused chatbot designed to detect mood patterns and depressive symptoms through pre-set questions and responses. This AI-driven tool engages users in natural conversations, gathering meaningful insights to assist therapists in assessing individual mental states. By automating preliminary evaluations, the chatbot enables mental health professionals to prioritize critical cases while providing timely support to all users. This solution delivers a private, scalable, and easily accessible method for early mental health screening particularly valuable in remote or underserved areas with limited psychological care resources. The facial analysis powered virtual healthcare system represents a groundbreaking resource for psychological practitioners, merging artificial intelligence- powered mood detection, speech evaluation, and interactive Chatbot features to create a holistic and accessible mental health tracking system. This technological synthesis improves evaluation precision

while encouraging preventative mental healthcare strategies, leading to improved treatment results for individuals.

1.2 DEEP LEARNING

Deep learning involves computational models with multiple processing layers that learn hierarchical representations of data. These advanced techniques have significantly enhanced performance across various fields, including speech recognition, computer vision, medical research, and genomics. By leveraging backpropagation, deep learning models automatically adjust their internal parameters to extract increasingly complex features from raw data. Convolutional neural networks (CNNs) have revolutionized image, video, and audio processing, while recurrent neural networks (RNNs) excel in analyzing sequential data like text and speech. Modern society relies heavily on machine learning for applications ranging from web search algorithms and social media content filtering to personalized recommendations on e-commerce platforms. These systems are now embedded in everyday devices, including digital cameras and smartphones, performing tasks such as image classification, speech-to-text conversion, content personalization, and search optimization. A key driver behind these advancements is deep learning, which overcomes the limitations of traditional machine learning methods in processing unstructured, natural data.

1.2.1 IMAGE RECOGNITION

Images are facts that depict visual perception. They have been used for diagnosis and detection in various fields including medical, agricultural and other fields. There are different existing image datasets that are on the cloud accessed when training different models. Various computer vision studies have used image data sets in classification, detection, recognition and segmentation of different research problems. Recently, the character portrayal is commonly used in disease diagnostics using images. Different levels of features are captured and analyzed based on various aspects, including color. Images are pre-processed (labeled and tuned) in order to maintain the realization of the former facts.

1.2.2 CONVOLUTIONAL NEURAL NETWORKS

Convolutional Neural Network (CNN) is a specialized type of deep learning model designed to process and interpret visual inputs, such as images. Its structure is built around convolutional layers that apply filters to scan input images and capture significant features like edges, textures and shapes. These features are then refined through the pooling layers, which compress

the data while retaining critical information, helping to reduce computational load. After feature extraction, the data flows through fully connected layers that analyze and combine the extracted patterns to make predictions or classifications. CNNs are trained on labeled image datasets, enabling them to learn and associate specific visual characteristics with certain objects or categories. Thanks to their layered design and capability to understand spatial hierarchies in data, CNNs are highly effective in applications such as image recognition, object detection, and face identification. They are widely used in today's computer vision systems due to their accuracy and efficiency.

1.2.3 SUPERVISED LEARNING

Supervised learning is a machine learning paradigm where an algorithm learns from labeled data, consisting of input-output pairs, to make predictions or decisions. In supervised learning, the algorithm is provided with a dataset containing input features and their corresponding labels or target values. The algorithm to learn a mapping or relationship between the input features and the output labels, so that it can generalize and accurately predict the output for new, unseen input data.

1.2.4 BACK PROPAGATION TO TRAIN MULTILAYER ARCHITECTURES

From the earliest days of pattern recognition, the aim of researchers has been to replace and engineered features with train able multilayer networks, but despite its simplicity, the solution was not widely understood until the mid-1980s. As it turns out, multilayer architectures can be trained by simple stochastic gradient descent. As long as the modules are relatively smooth functions of their inputs and of their internal weights, one can compute gradients using the back propagation procedure. The idea that this could be done, and that it worked, was discovered independently by several different groups during the 1970s and 1980s.

The back propagation procedure to compute the gradient to an objective function with respect to the weights of a multilayer stack of modules is nothing more than a practical application of the chain rule for derivatives. The key insight is that the derivative (or gradient) of the objective with respect to the input of a module can be computed by working backwards from the gradient with respect to the output of that module (or the input of the subsequent module). The back propagation equation can be applied repeatedly to propagate gradients through all modules, starting from the output at the top (where the network produces its prediction) all the way to

the bottom (where the external input is fed). Once these gradients have been computed, it is straight forward to compute the gradients with respect to the weights of each module.

1.3 PROPOSED METHOD

The proposed project offers a virtual healthcare platform aimed at evaluating mental health by combining facial emotion detection and voice-based sentiment analysis. Counsellors access the system through a secure login, after which facial emotions are analyzed in real-time using Convolutional Neural Networks (CNN) with TensorFlow and OpenCV. Spoken inputs are processed using the Speech Recognition library and further analyzed with the TF-IDF algorithm to determine emotional tone. This multi-modal approach improves the system's ability to accurately detect signs of conditions like depression. Based on the evaluation, the application delivers personalized suggestions, such as prompts for meditation. A chatbot feature is included to simulate counsellor-like conversations through preset questions and responses. User data is securely stored and managed via SQLite Studio for convenient tracking. For data analysis and visualization, Pandas and Matplotlib are used, while Flask serves as the backend to ensure a responsive web interface. Altogether, the system offers an intelligent and accessible solution for remote mental health and support.

1.4 LITERATURE SURVEY

Archana Sharma et al. [1] introduced a multi-modal system combining facial expressions, voice, and text inputs with deep learning techniques to detect depressive symptoms and enhance emotional analysis through a supportive chatbot. Jiawei Yu et al. [2] focused on emotionally charged speech using an emotion-based model to identify depression-related vocal segments, emphasizing the value of emotional tones in mental health analysis. Devesh Shukla et al. [3] presented a deep learning-based method integrating physiological ECG signals and emotional text data for effective emotional fatigue detection. Nirupa V et al. [4] developed an emotion-aware chatbot using facial expression recognition via a CNN model, enabling more emotionally adaptive and human-like interactions. Farshad Safavi et al. [5] proposed a deep multimodal network that combines neurophysiological and facial features to improve emotion recognition accuracy, advancing emotion-aware technology in human- computer interaction.

A. Sharma et al. [6] reviewed recent advancements in real-time emotion detection, focusing on the use of machine learning algorithms for analyzing facial expressions and emphasizing the need for accurate and efficient methods. P. Mishra et al. [7] investigated deep learning models

such as CNN, VGG16, and VGG19 for facial emotion recognition, stressing the improvements in classification performance through advanced architectures. Pushpavalli Mohan et al. [8] developed a lightweight deep learning system that combines dynamic facial analysis with attention mechanisms to improve the precision and clarity of emotion recognition. Jingyi Yuan et al. [9] introduced a GRU-based model that identifies depressive tendencies by analyzing patterns in speech and text, illustrating the benefits of a multimodal approach. Arya et al. [10] proposed a machine learning solution for predicting depression, integrating traditional algorithms with data enhancement techniques to strengthen mental health evaluations across various datasets.

Wentao Yan et al. [11] proposed an enhanced text similarity method by combining TF-IDF with Lex Rank, improving the accuracy of semantic understanding in text analysis. Shuvra Tripura et al. [12] emphasized the role of effective software architecture in building high-performing image recognition systems, utilizing machine learning tools to streamline data processing and improve model outcomes. Ms. Priyanka Wani et al. [13] developed an Automatic Voice Recognition (AVR) system designed to assist individuals with disabilities by converting speech into text using machine learning techniques. Jianqiang Xu et al. [14] introduced a multi-modal deep learning approach that integrates physiological signals and emotional text data to enhance emotional fatigue detection. Ezerceli et al. [15] advanced facial emotion recognition by applying convolutional neural networks, demonstrating the effectiveness of modern deep learning models in improving emotion classification accuracy.

Ansari Abu Affan Mustafa Kamal Ah et al. [16] Ansari Abu Affan Mustafa Kamal Ah et al. [16] proposed a facial emotion detection system using CNNs to classify emotions and a recommendation framework that suggests personalized motivational videos based on detected emotions. Arnav Kotiyal et al. [17] explored the performance of six pre-trained CNN models for image prediction tasks, with EfficientNetB7 standing out for its high accuracy, while also demonstrating the advantages of transfer learning. Wu et al. [18] developed an end-to-end deep learning framework for real-time face detection and facial landmark localization, significantly improving efficiency and accuracy in facial analysis. Suryamritha et al. [19] introduced a CNN-LSTM hybrid model for speaker identification, achieving high accuracy and emphasizing the importance of combining spatial and temporal features in audio-based tasks. Shivakumar et al. [20] advanced automatic speech recognition by integrating multi- modal large language models (LLMs) to combine speech and text, greatly improving model performance and reducing word error rates.

Garima Chandel et al. [21] developed a healthcare chatbot that uses AI, deep learning, and natural language processing (NLP) to provide fast, personalized medical assistance, offering symptom analysis and basic treatment advice while easing the burden on healthcare professionals. Varshaa Dhanasekar et al. [22] created Maxx, a chatbot designed to support students' mental health by identifying emotions and providing tailored guidance on issues like stress, loneliness, and academic pressure, especially during the pandemic. Zhou Kun et al. [23] introduced a speech synthesis system capable of expressing mixed emotions, enhancing the naturalness of speech with nuanced emotional expressions, though challenges remain in cultural adaptation and natural blending. Boumahdi Fatima et al. [24] proposed a deep learning model that detects early signs of depression by analyzing social media posts, achieving high accuracy with a hybrid CNN-BiLSTM model to enhance early depression detection. S. Abinaya et al. [25] developed TEBC-Net, a chatbot combining text and facial emotion analysis to create more empathetic conversations, using a fine-tuned BERT model and a CNN model for emotion detection, with potential applications in mental health and customer service.

Table 1.1 illustrates a clear overview of selected research papers. It includes the reference names, datasets used, methods or results, and any limitations mentioned in the studies. This summary helps to show common patterns, strengths, and gaps in the existing research related to this topic.

Table 1.1: Comparative Analysis of Related Research Works

SL No.	Reference Paper	Dataset	Methods / Results	Limitation
1.	Archana Sharma et al. “Detection using multimodal analysis with chatbot support” IEEE, 2024.	Custom dataset (facial, audio, text via chatbot)	Multimodal approach to improve detection accuracy and chatbot to enhance user engagement.	Small dataset, limited generalizability for emotion recognition, and chatbot not a substitute for clinical diagnosis.
2.	Jiawei Yu et al. “Using Emotionally Rich Speech Segments for Depression Prediction.” IEEE, 2025.	DAIC-WOZ and CMDC datasets	Using emotionally rich speech, improved depression prediction accuracy to 79% F1 on DAIC-WOZ and 90.53% on CMDC.	Limited generalizability due to dataset constraints and potential challenges in real-world applicability.

3.	Shukla et al. and Emotion Recognition Using OpenCV through AI." IEEE, 2024.	Likely custom or real-time webcam data is used	Utilized OpenCV for real-time face detection and emotion recognition, demonstrated effectiveness in controlled environments.	Lack of publicly available dataset, limited testing on diverse populations, and potential challenges in varying lighting conditions and facial orientations.
4.	Nirupa V et al. "Chatbot Based on Emotions Using Deep Learning." IEEE, 2023	FER 2013 (Kaggle) + Custom Dataset (2000+ images)	The study achieved 85%+ accuracy in emotion detection using CNN (VGG-Face), with tailored chatbot responses and improved adaptability.	The study's limitations include a small custom dataset 2000 images restricting generalization, reliance on static images without dynamic emotion tracking.
5.	Farshad Safaviet et al. "Deep Fusion of Neurophysiological and Facial Features for Enhanced Emotion Detection." IEEE, 2025	DEAP and LieWaves datasets	The model achieved over 97% accuracy by fusing facial and EEG features.	Variability in EEG signals and limited datasets affect performance.
6.	A. Sharma et al. "Machine learning techniques for real-time emotion detection from facial expressions." IEEE, 2023	Standard datasets like CK+, JAFFE, FER2013	The paper compares ML methods for emotion detection, with Random Forest (97.17%) CNN/Eigenface (~96%) achieving top accuracy.	No dataset specifics; unaddressed challenges (lighting, occlusion, aging); lacks experimental validation on unified data.
7.	P. Mishra et al. "Emotion Recognition from Facial Expression Using Deep Learning Techniques" IEEE, 2024	FER2013 Dataset	The study compares VGG16, VGG19, and a custom CNN for facial emotion recognition, with VGG19 achieving 92.5% accuracy on the FER2013 dataset.	The FER2013 dataset lacks diversity, VGG19's high computational cost limits real-world use, and newer models weren't compared. These factors affect generalizability and practical application.

8.	Pushpavalli Mohan et al. “Deep Learning Based on Face Emotion Recognition using an Artificial Neural Network” IEEE, 2023	FER2013 and CK+ datasets	Deep learning models achieve higher accuracy on CK+ (97.5%) than FER2013(70.13%) MobileNetV2 with accurate real-time emotion recognition.	MobileNetV2 performs well on CK+ (97.5%) but poorly on FER2013 (70.13%), revealing real-world limitations.
9.	Jingyi Yuan et al. “Depressive Tendency Recognition Using the Gated Recurrent Unit from Speech and Text Features” IEEE, 2021	Depression Corpus and Chinese Facial Expression Picture System datasets	The GRU model effectively detects depression better via speech (73%) than text (65.9%), with higher accuracy in females (78%) than males (69.9%).	The GRU model achieves 73% depression detection via speech (vs 65.9% text), with 78% female accuracy (69.9% male).
10.	Arya et al. "Predicting Depression and Mental Illness Using Machine Learning Algorithms." IEEE, 2023	The Kaggle Depression Dataset (D1) has 1075 entries and 12 features, while the Survey Dataset (D2) has 269 entries with 15 features	Support Vector Machine performed best on Dataset 1 without SMOTE, while Random Forest with SMOTE and Decision Tree achieved 100% on Dataset 2.	The small size of Dataset 2 (269 records) increases over fitting risk, evident from the Decision Tree's 100% accuracy, and limits real-world validation.
11.	Wentao Yan et al. “An Improved Text Similarity Calculation Method Combining TF-IDF and LexRank" IEEE, 2024	DUC dataset, Chinese text datasets, Scientific research project texts	The combined TF-IDF and LexRank method showed higher accuracy and F1-scores.	The method increased computational complexity and resource requirements.
12.	Tripura et al. "Architecture of Data Recognition System Using Machine Learning and TensorFlow" IEEE, 2024	Celebrity image dataset	The system achieved over 90% classification accuracy after training using TensorFlow and API-based data fetching.	High training time depending on device performance despite faster API-based data handling.

13.	Ms. Priyanka Wani et al. "Automatic Voice Recognition System" IEEE, 2024	Hindi Utterances (Regional Indian Accents - RIA dataset)	The system using MFCC and classifiers like HMM, KNN, and ANN achieved nearly 90% recognition accuracy.	Recognition accuracy is challenged by speaker variability, background noise, and device transmission effects.
14.	Jianqiang Xu et al. "Intelligent emotion detection method based on deep learning in medical and health data" IEEE, 2020	ECG data and WeChat emotional text dataset	The proposed multi-modal deep learning model achieved over 85% average accuracy in emotional fatigue detection.	Performance varied due to sample size differences and subjective emotional labelling.
15.	Ezerceli et al. "Convolutional neural network (CNN) algorithm based facial emotion recognition (FER) system for FER-2013 dataset" IEEE, 2022	FER2013, CK+, JAFFE, FERG, MMI	The model achieved 93.7% accuracy using combined FER2013 and CK+ datasets.	Performance was affected by occlusion, pose, resolution, and class imbalance.
16.	Ansari Abu Affan Mustafa Kamal Ah et al. "Facial Emotion Detection and Recommendation using Deep learning" IEEE, 2024	FER2013	The CNN model trained on FER2013 accurately detects seven emotions and recommends motivational content.	Performance may vary due to factors like real-time video quality, facial occlusions, and lighting conditions.
17.	Arnav Kotiyal et al. "Pre-trained Model for Image Prediction Using Keras Application" IEEE, 2023	Pre-trained ImageNet weights Efficient Net, InceptionV3, MobileNet, MobileNetV 2, Xception, ResNet50	Comparative study of six Keras-based pre-trained models for image prediction; EfficientNetB7 achieved highest accuracy (84%–97%).	Limited to pre-trained ImageNet datasets; no new datasets or fine-tuning applied; lacks domain-specific evaluation.
18.	Wu et al. "Real-Time Face and Facial Landmark Joint Detection Based on End-to-End Deep Network." IEEE, 2025.	300W, COFW, AFLW2000-3D datasets	Proposed an end-to-end YOLO-based network for real-time face and landmark achieving 102 FPS with robust performance across datasets.	Focused mainly on 68-point dense landmarks; struggles under severe rotations and partial occlusions; relies heavily on YOLO adaptations.

19.	Suryamritha et al. "Speaker Identification Using CNN-LSTM Model on RAVDESS Dataset: A Deep Learning Approach." IEEE, 2024.	RAVDESS dataset	Proposed a hybrid CNN+LSTM model combining spatial and temporal features for speaker identification, achieving 96.52% accuracy and 97% F1-score.	Model performance is validated only on a single emotional dataset (RAVDESS), limiting broader generalization across languages and demographics.
20.	Gurunath Shivakumar et al. "Speech Recognition Rescoring with Large Speech-Text Foundation Models" IEEE, 2025.	LibriSpeech, LibriLight, CoVOST2, TED-LIUM-3, People Speech, Internal datasets	Proposed multi-modal LLM-based ASR rescoring improve WER by up to 20% over Whisper large ASR demonstrated cross-modal learning advantages.	Requires large computational resources; effectiveness relies on extensive multi-modal pretraining; limited exploration on low-resource domains.
21.	Garima Chandel et al. "Healthcare Chatbot for Personal Healthcare Assistance." IEEE, 2024.	Chatbot models trained using medical symptom datasets and simulated patient interaction	Developed a chatbot using Python NLP libraries with five classifiers: SVM, KNN, Naive Bayes, Decision Tree, and Random Forest, which achieved 98.43% accuracy.	Real clinical testing and benchmark datasets were not included, raising concerns about over reliance on automation, which cannot replace medical expert judgment.
22.	Varshaa Dhanasekar et al. "A chatbot to promote students mental health through emotion recognition." IEEE, 2021	Positive emotion data was from DialogFlow and negative emotion data from government and UN websites	Developed Maxx chatbot using Dialogflow, Flutter, and Google Cloud to detect emotions and give advice, achieving 90% accuracy for positive and 80% for negative emotions.	Limited data for emotions like jealousy and ego may cause chatbot errors with complex issues. No real-time psychiatric feedback unless manually requested.
23.	Zhou Kun et al."Speech Synthesis with Mixed Emotions." IEEE, 2023.	VCTK Corpus (multi-speaker) and ESD (Emotional Speech Dataset)	Proposed a relative ranking sequence-to-sequence TTS framework to synthesize speech with mixed emotions, validated through Objective and	Initial focus only on English speakers, challenges with extreme emotion blending, and computational complexity of emotion control.

			subjective evaluations.	
24.	Boumahdi Fatima et al. "A mixed deep learning-based model to early detection of depression." Journal of Web Engineering, 2020.	eRisk 2018 dataset (Reddit posts)	Proposed a CNN-BiLSTM with Attention model achieving 97% accuracy for early depression detection from social media text data.	Focused mainly on English Reddit posts; challenges in generalizing across languages, cultures, and smaller social media datasets.
25.	S. Abinaya et al."Enhanced Emotion-Aware Conversational Agent: Analysing User Behavioural Status for Tailored Reponses in Chatbot Interactions". IEEE 2025	Combined Daily Dialog, ISEAR, Emotion-Stimulus datasets (text) + FER2013 (images)	Developed TEBC-Net, a multi-modal chatbot combining text (BERT) and video (CNN) emotion recognition, achieving 87.21% (text) and 74.14% (video) accuracy.	Focused on English-only datasets; privacy concerns in real-time video analysis; generalization to diverse demographic settings not fully explored.

REQUIREMENT SPECIFICATION AND ANALYSIS

CHAPTER 2

REQUIREMENT SPECIFICATION AND ANALYSIS

2.1 INTRODUCTION

Requirement analysis is the process of creating a score for a system's effort and is critical to the success or failure of a system or software project. The requirements should be documented, actionable, measurable, testable, traceable, and linked to identified business needs or opportunities, defined to a level of detail sufficient for system design. It helps stakeholders understand what needs to be built and ensures that all expectations are aligned before development begins. Thorough requirement analysis reduces the risk of project failure by identifying potential issues early in the development lifecycle.

2.2 FUNCTIONAL REQUIREMENTS

In software engineering and systems engineering, a functional requirement defines a function of a system or its component, where a function is described as a specification of behavior between inputs and outputs.

2.2.1 USER REQUIREMENTS

- User should be able to access the application in laptop using webcam.
- User should be able to give input as face for detection.
- User’s face must be recognized by certain emotions.
- User should be able to interact with the Voice Assistant.
- User should be able to use the Chatbot for Q&As.

2.2.2 SYSTEM REQUIREMENTS

- The system should be able to capture an image input.
- The system should provide an user interface in the form of a web application to allow the user to predict mental illness such as depression.
- The system should securely store user data by including facial recognition data and emotion analysis results.

- The system should be able to provide communication via a Chatbot.

2.3 NON-FUNCTIONAL REQUIREMENTS

The system shall ensure that facial analysis results are generated within 3 seconds with an accuracy rate of at least 95%, while maintaining strict data privacy and ensuring a user-friendly interface accessible on both mobile and desktop platforms. Additionally, the system shall support at least 500 concurrent users without performance degradation and ensure 99.9% uptime to provide continuous healthcare services.

2.3.1 Reliability

To ensure the reliability of the Facial Analysis Powered Virtual Healthcare system, it must be capable of handling unexpected shutdowns, network disruptions, and hardware failures without compromising the integrity of user data.

2.3.2 Accuracy

The facial analysis and emotion detection algorithm should achieve at least >85% accuracy to ensure reliable health assessments. The system must be tested with diverse datasets to minimize biases across different age groups, genders, and ethnicities. Continuous learning and model retraining should be employed to improve detection precision over time

2.3.3 Privacy

The system must ensure secure data storage and handling in compliance with GDPR regulations to protect user privacy and confidentiality. User data should be anonymized wherever possible to prevent identification. Access controls, data encryption (both in transit and at rest), and regular audits should be implemented to maintain high levels of security.

2.3.4 Performance

The system should deliver real-time responses with a delay of less than 2 seconds to ensure smooth and efficient user interaction. Load testing should be conducted to validate performance under high user traffic.

2.3.5 Usability

The user interface should be easy to navigate, catering to both tech-savvy and non-technical users. The system should provide clear and concise instructions to users, with tooltips and help sections readily available.

2.3.6 Scalability

The system should be able to handle an increasing number of users and data without degradation. It should support horizontal and vertical scaling to manage growing demands. Cloud infrastructure and containerization can be leveraged to facilitate seamless expansion and deployment.

2.3.7 Maintainability

Proper documentation should be provided for all system components, including APIs, databases, and algorithms. The system should allow easy updates and maintenance without disrupting ongoing services.

2.4 SOFTWARE REQUIREMENTS

The project requires following software to run.

- Operating System – 64-bit Microsoft® Windows® 8/10 or MacOS 10.
- Languages – Python, XML.
- IDE – Visual Studio Code.
- Python Packages–Matplotlib, Flask, OpenCV, TensorFlow, Keras, Speech recognition.

2.4.1 OpenCV

In the facial analysis powered virtual healthcare system, OpenCV plays a critical role in capturing and processing real-time facial data from video input. These facial landmarks are then passed to deep learning models integrated through libraries such as TensorFlow, Keras for emotion classification and health-related analysis.

2.4.2 DeepFace

Deepface is used for real-time facial emotion detection from live video feeds. Integrated with TensorFlow and Keras, it accurately classifies emotions like happiness, sadness, and stress. These insights enable the system to deliver personalized, empathetic healthcare responses.

2.4.3 TensorFlow

In the virtual healthcare system, TensorFlow powers deep learning models for real-time facial emotion recognition, enabling accurate emotion prediction from expressions.

2.4.4 Pillow

Pillow is used in the project for handling and preprocessing image data captured from user video

input. It allows for image resizing, format conversion, and enhancement techniques such as blurring or edge detection to improve facial feature extraction accuracy.

2.4.5 Flask

Flask serves as the web framework that powers the front-end and back-end integration of the healthcare system. It manages HTTP requests, connects the user interface with emotion detection models, and delivers real-time feedback and healthcare responses based on user interactions.

2.4.6 Matplotlib

In this project, Matplotlib is used to generate visual reports that represent emotion trends over time. These visualizations help both users and healthcare providers understand behavioral patterns and emotional states during virtual consultations.

2.4.7 NumPy

NumPy is essential for processing image and video data by converting visual inputs into numerical arrays. These arrays enable deep learning models to accurately and efficiently detect facial emotions.

2.4.8 Speech Recognition

The Speech Recognition library in Python enables the conversion of spoken input into written text by interfacing with various engines like Google Web Speech API. After transforming speech to text, the system employs TF-IDF (Term Frequency–Inverse Document Frequency) to extract meaningful features by assigning weight to words based on their relevance across multiple texts. These feature vectors are then passed to a Multinomial Naive Bayes classifier, which categorizes the input based on underlying emotional content.

2.4.9 TF-IDF Vectors

TF-IDF is a Natural Language Processing technique that converts text into numbers based on a word's importance within a document and its relevance across a corpus.

2.5 HARDWARE REQUIREMENTS

- RAM:8GB
- Storage: Minimum requirement of 256 GB SSD
- Camera: A high-definition webcam (1080p or higher)
- Intel Core Processor (i3 and above)

SYSTEM DESIGN

CHAPTER 3

SYSTEM DESIGN

3.1 INTRODUCTION

The system design section provides a comprehensive overview of the architecture, modules, and key components of the system. It aims to describe the structure and design decisions made to ensure the efficiency and effectiveness of the AI-Based Depression Detection System. The system is developed to provide a supportive platform for individuals facing emotional distress or depression. By integrating facial emotion recognition technology with an AI chatbot, the system helps users assess their emotional state and receive personalized support. The design of the system focuses on creating an intuitive, user-friendly interface, reliable emotion detection, and seamless chatbot interaction. This section outlines the rationale behind the design choices and the architecture that enables smooth performance and interaction. Each component is carefully considered to work together in a way that offers an optimal user experience while maintaining system reliability and accuracy.

3.2 PURPOSE AND SCOPE OF DESIGN

The primary objective of this system design is to provide a comprehensive and structured overview of the AI-Based Depression Detection System, which is developed to assist users in identifying their emotional states through either facial recognition or interaction with an intelligent chatbot. The system emphasizes a user-friendly and intuitive interface to ensure seamless navigation, particularly for individuals experiencing emotional distress. At its core, it features a real-time emotion detection module utilizing advanced machine learning techniques to accurately classify emotions based on facial expressions. Additionally, a responsive chatbot offers immediate emotional support, adapting its responses based on detected emotions and recommending appropriate coping strategies when needed. All components are integrated within a strong backend to ensure smooth performance, quick response times, and secure data handling, providing a dependable solution that supports emotional well-being. This system is designed to be accessible and user-friendly, making it easier for individuals to seek help anytime. Continuous updates and testing ensure that the platform remains reliable and aligned with user needs. The system also prioritizes user privacy, ensuring all emotional data is handled with strict confidentiality and security measures.

3.3 IMPORTANCE OF DESIGN DECISIONS

The design choices made for the AI-Based Depression Detection System are crucial to its success and functionality. Key factors influencing the design include:

- **User Experience (UX):** A simple, clean interface allows users to easily navigate the system. Ensuring the system is easy to use is a primary goal, as this increases engagement and satisfaction for users who may be experiencing emotional distress.
- **Accuracy of Emotion Detection:** The system utilizes state-of-the-art AI models, particularly Convolutional Neural Networks (CNN), to accurately classify emotional states. This ensures that users receive precise feedback based on their emotional expressions, which is essential for providing appropriate responses.
- **Scalability and Flexibility:** The system is built to be scalable, allowing it to grow as new features or improvements are introduced. This flexibility ensures that the system can be updated to address emerging needs or integrate with other technologies or services.

3.4 SYSTEM ARCHITECTURE DESIGN

The system is divided into several key components, each responsible for handling a specific function. These components work collaboratively to ensure smooth operation and deliver a seamless user experience. The main components include:

- **User Interface (UI):** The UI serves as the primary interaction point for users, allowing them to engage with the system. It includes the registration and login pages, the emotion capture interface, and the chat window for interaction with the chatbot.
- **Emotion Detection Module:** This module uses facial recognition algorithms to detect the user's emotional state through real-time video input. Once the emotion is detected, the system processes the data and forwards it to the chatbot for response generation.
- **Chatbot Integration:** The chatbot module uses the emotion data to formulate appropriate, empathetic responses to users. It engages in supportive dialogue with the user, providing emotional assistance based on the detected emotional state.
- **Backend System:** The backend system acts as the central processing unit of the platform. It handles the communication between the UI, emotion detection module, and chatbot, ensuring that data flows smoothly throughout the system. It also stores data and session histories for continuity in interactions.

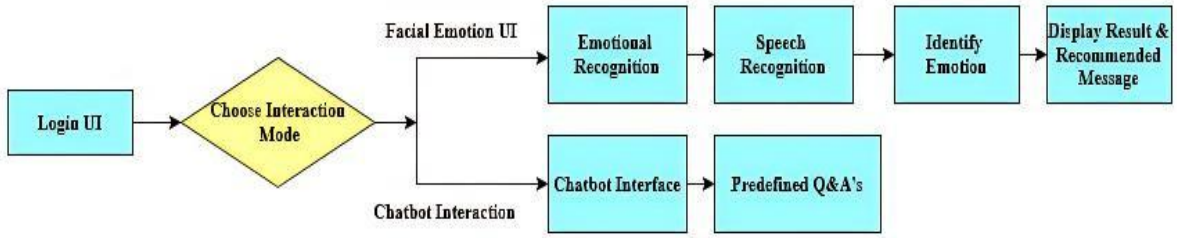

Figure 3.1: Architecture of the System

The above figure 3.1 illustrates the system architecture of the depression detection and support system. After logging in, the user selects an interaction mode either facial emotion recognition or chatbot interaction. In the facial emotion path, the system captures facial expressions and speech input to recognize and identify emotions, then displays the result along with supportive messages. In the chatbot path, the user interacts with a predefined question-and-answer-based chatbot that responds empathetically based on user input. This structure ensures a user-friendly, flexible, and emotionally intelligent system.

3.5 FLOW CHART

A flowchart is a visual representation of a process, system, or algorithm that uses standardized symbols to depict the sequence of steps, decisions, and actions. It serves as a valuable tool in both planning and documentation, enabling clear communication of complex workflows in a simplified and structured manner. In the context of this project, the flowchart outlines the operational flow of an intelligent emotion-aware system that integrates facial analysis and chatbot functionalities. The primary objective of this system is to recognize or receive input about a user’s emotional state and respond with appropriate feedback or support. The project aims to enhance user interaction and emotional engagement through two key modules: one that utilizes real-time facial emotion recognition using camera and speech input, and another that allows users to manually select emotions to interact with a predefined chatbot. The flowchart provides a step-by-step visualization of the system's functionality, beginning with user registration and guiding through both modes of interaction, ultimately delivering personalized suggestions or responses based on the identified or selected emotion. This clear visual representation ensures a better understanding of the system's workflow, aiding both development and future enhancements.

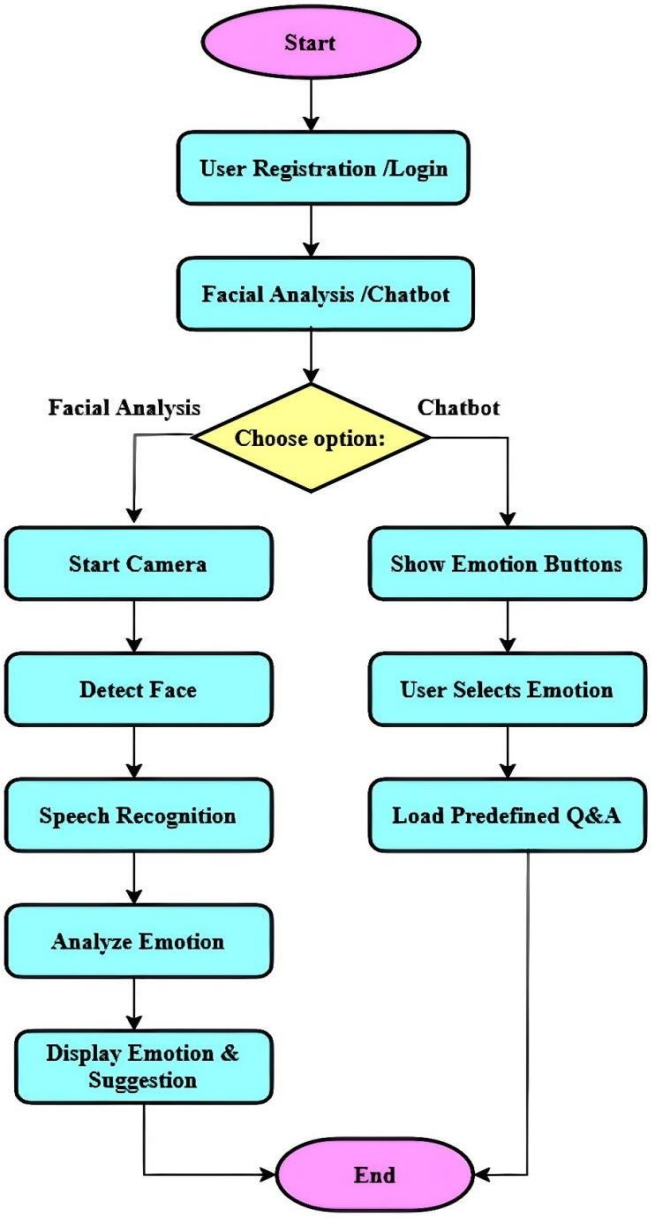

Figure 3.2: Flowchart of the Project

In the Figure 3.2 flowchart represents the workflow of a system designed to offer two functionalities: facial emotion analysis and a chatbot-based interaction. The process begins with the user registration or login, after which the user is presented with two options either to proceed with facial analysis or interact with the chatbot. If the user selects the facial analysis option, the system initiates the camera to capture the user's face. It then detects the face and performs speech recognition to gather both visual and audio cues. These inputs are analyzed to determine the user's emotional state, and based on the analysis, the system displays the detected emotion along with relevant suggestions or recommendations. On the other hand, if the user opts for the chatbot, the system displays a set of emotion buttons for the user to select how they are feeling.

Once an emotion is selected, the system loads a set of predefined questions and answers tailored to the chosen emotion to offer a supportive conversational experience. The process concludes after either emotion-based suggestions are displayed or the chatbot interaction is completed. This system is intended to provide an emotion-aware experience, making it useful for mental health support, personalized AI interaction, or emotional well-being monitoring.

3.6 USE CASE DIAGRAM

Use case diagrams provide a clear and structured visual representation of the interactions between various actors such as users or external systems and the system itself. These diagrams are fundamental in software design and system analysis, serving as essential tools for defining and communicating the functional requirements of a system. By offering a high-level overview, use case diagrams illustrate the services or actions the system delivers to its users and demonstrate how different components collaborate to fulfill these functionalities.

In the context of the AI-Based Depression Detection System, use case diagrams play a pivotal role in mapping the interactions between the user and the system. They outline how users engage with the system across key functionalities, including logging in, selecting between the emotion detection and chatbot modules, receiving tailored feedback based on the selected module, and concluding their session. Furthermore, these diagrams aid in understanding how the system processes user inputs, particularly through the integration of facial emotion analysis and chatbot-driven communication.

A typical use case diagram consists of the following core elements:

- **Actors**: Entities (individuals or external systems) that interact with the system.
- **Use Cases**: Specific functionalities or operations the system performs in response to actor inputs.
- **System Boundary**: A defined scope that encapsulates the use cases, indicating the operational limits of the system.
- **Associations**: Connective lines between actors and use cases that represent their interactions.

The below figure 3.3 illustrates how the user interacts with the system's key features. The user begins by logging in or registering and is then prompted to choose between emotion detection or chatbot interaction. If emotion detection is selected, the system captures the user's facial expression and provides feedback based on the detected emotion. If the chatbot option is chosen, the system processes the user’s text input and generates responses based on emotional context or the conversation. The user can end the session at any time. This diagram showcases the flow of

actions and responses between the user and the system.

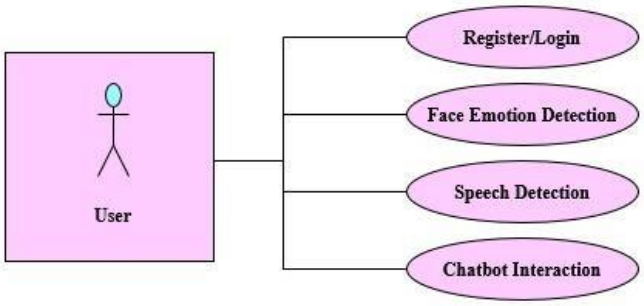

Figure 3.3: Use-case Diagram of the User

The figure 3.4 represents the architecture of the emotion detection system designed to process user inputs through multiple modalities. The system consists of components that handle facial emotion recognition, vocal emotion analysis, and textual input evaluation. The facial processing module captures and interprets visual cues from the user's expressions, while the vocal analysis component evaluates tone and speech patterns to identify emotional states. Simultaneously, the text analysis module examines written input to detect sentiment. These parallel processes contribute to a unified emotional assessment, which is then passed to the response generation module. Based on the detected emotional context, appropriate feedback is formulated and delivered to the user. This structure enables a more accurate and responsive system, supporting personalized and empathetic user interaction.

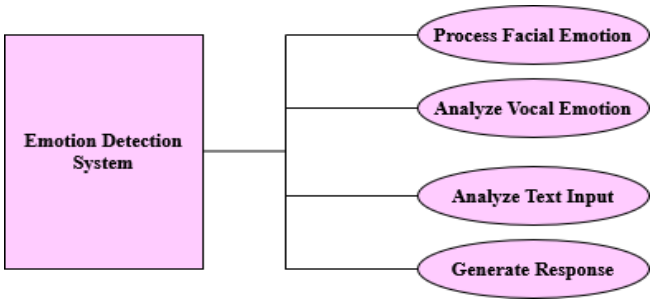

Figure 3.4: Use-case Diagram of the System

3.7 SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language is an interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios and timing

diagrams.

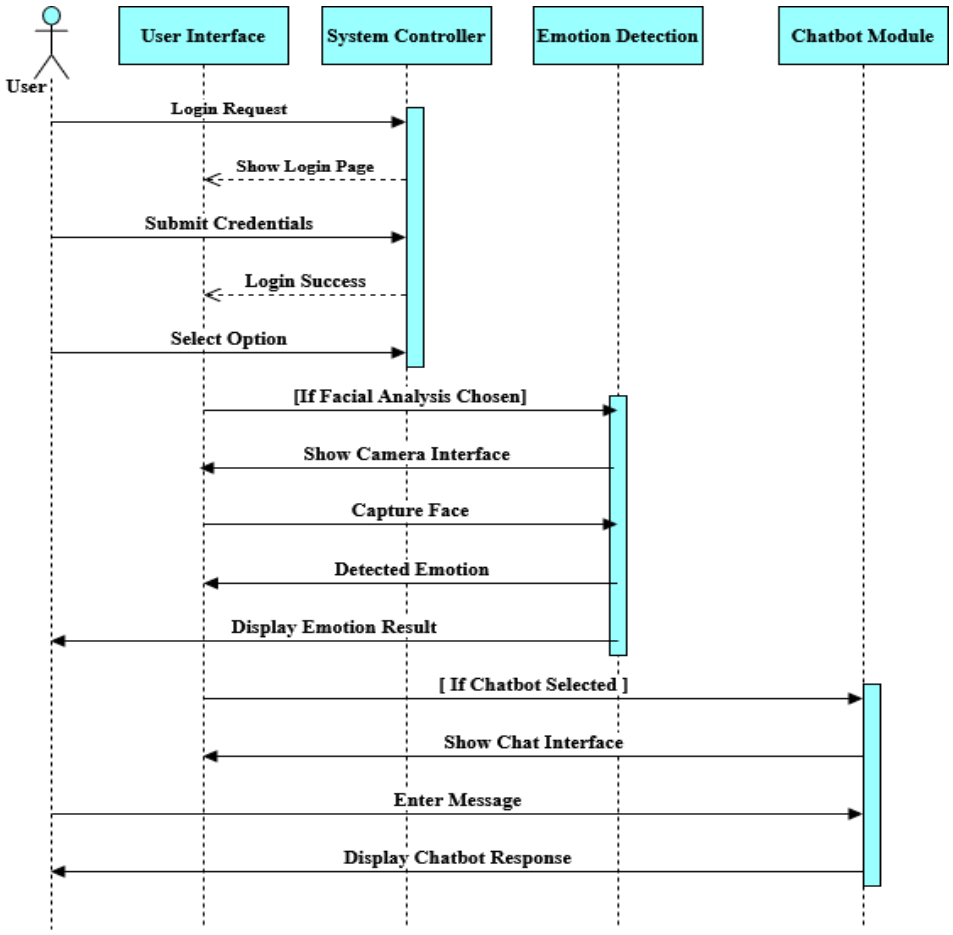

Figure 3.5: Sequence Diagram of the System

The above figure 3.5 illustrates the interactions between the user and the system, highlighting the flow of operations. The user begins by selecting the login option, prompting the User Interface to display the login page. After entering their credentials, the system authenticates the user and sends a confirmation of successful login back to the User Interface. Upon successful login, the user is presented with two options, emotion detection or chatbot interaction. If the user chooses emotion detection, the User Interface activates the camera to capture the user's facial expression. This image is then sent to the Emotion Detection module, where it is analyzed and classified. The system then returns the identified emotion to the User Interface, which displays the feedback to the user.

Alternatively, if the user selects chatbot interaction, the User Interface presents a chat interface. The user sends a message to the chatbot, which processes the input and generates an appropriate

response. This response is sent back to the User Interface and displayed to the user. This diagram clearly defines the sequence of interactions between the user, User Interface, system controller, emotion detection module, and chatbot module, ensuring a well-organized and efficient user experience.

SYSTEM IMPLEMENTATION

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

The Facial Analysis Powered Virtual Healthcare System is an AI-driven platform designed to identify early indicators of depression by evaluating facial expressions and speech patterns. Utilizing Convolutional Neural Networks (CNN) for real-time facial emotion recognition and TF-IDF (Term Frequency-Inverse Document Frequency) for analyzing user inputs in the chatbot module, the system delivers personalized mental health assessments. By integrating emotion detection with AI-powered conversational support, it helps users recognize emotional distress promptly and recommends tailored interventions such as mindfulness exercises or referrals to professional counselling. This innovative solution merges advanced technology with mental health care, offering an accessible, proactive approach to emotional well-being.

4.2 PROJECT WORKFLOW

The system workflow begins with:

- **Web Page (UI) –** Interface for counsellors to monitor patients.
- **Login/Registration (Counselors Only) –** Mental health professionals' register/login to access patient data.
- **Start Analysis –** Counselor initiates emotion detection for a patient.
- **Detect Face Emotion –** Patient's facial expressions analyzed via CNN.
- **Speech Recognition –** Patient's voice assessed for emotional tone.
- **Result & Recommendation –** System provides insights (e.g., "depression detected – Suggest therapy session").
- **Chatbot Application –** AI chatbot assists counselors with preliminary patient interactions.

4.3 DATASET COLLECTION

The system leverages publicly available benchmark datasets to train its AI models for accurate emotion recognition. For facial expression analysis, we utilize the FER-2013 dataset These datasets enable our CNN model to detect subtle emotional cues from facial features.

For speech emotion recognition, the system captures voice input in real time and converts it into text using an integrated speech recognition module. After transcription, the text is analyzed alongside the tweet_emotion.csv dataset to identify emotional patterns and sentiments. To continuously improve the model's performance, the system securely stores user voice inputs and chatbot conversations for future retraining. It also supports document uploads, automatically extracting and organizing content into a structured CSV format for easy data integration. With these combined features, the system ensures accurate emotion detection while streamlining data preparation and analysis.

4.4 DATA PREPROCESSING

The system follows a multi-step approach to preprocess data, ensuring optimal model performance for various input types. For facial recognition, images are resized, normalized, and enhanced through techniques like rotation and flipping to help the CNN model generalize more effectively. Within the CNN, the Convolutional2D layer identifies key features such as edges, while the ReLU activation function filters out negative values. The MaxPooling2D layer reduces the dimensionality, and the Dropout layer minimizes overfitting by randomly disabling certain neurons. After that, the Flatten layer transforms the data for processing, while the Dense layers interpret the information, and the SoftMax layer determines the predicted emotion. The minNeighbours parameter in the multiscale function aids in identifying the most common emotion in the image. For speech emotion recognition, raw audio is transcribed into text using Python's Speech Recognition library, which is then processed using TF-IDF vectorization and classified with a Multinomial Naive Bayes (MultinomialNB) pipeline. For text interactions in the chatbot, user inputs are tokenized, stop words are removed, and punctuation is stripped before being converted into TF-IDF vectors. This integrated preprocessing workflow ensures that the data is clean and structured, enabling precise emotion detection from facial, vocal, and text inputs.

4.5 MODEL EVALUATION ALGORITHM

The system employs two distinct machine learning techniques to achieve a comprehensive analysis of emotional states. the system uses the Haar Cascade algorithm for reliable face detection, ensuring that facial features are accurately located before any further analysis. It then applies two separate machine learning methods to interpret emotional states from different inputs. For facial emotion detection, a Convolutional Neural Network (CNN) is used, which captures detailed facial features through several layers of convolution. These features are then

classified into emotions like happy, sad, angry, or neutral using a SoftMax activation function, with backpropagation used during training to improve prediction accuracy. In parallel, speech emotion recognition is handled by converting spoken audio into text, transforming the text using TF-IDF vectorization, and then feeding it into a Multinomial Naive Bayes model to determine the emotional tone behind the speech. Additionally, the chatbot component processes user text inputs using the same TF-IDF technique and matches them to the most suitable counsellor responses through cosine similarity. This combined approach covering facial cues, voice analysis, and text interpretation enables the system to detect emotional states in real time while offering deeper contextual understanding. All models are carefully tested using accuracy, precision, and recall metrics to ensure dependable performance before deployment.

4.6 MODEL TRAINING AND VALIDATION

The system's model training and validation pipeline leverages a comprehensive tech stack including Python, TensorFlow, Kera's, OpenCV, and scikit-learn for robust AI development. For facial emotion recognition, the CNN model is trained on labeled datasets using TensorFlow/Kera's, with OpenCV handling real-time face detection and processing (resizing, normalization). Training metrics (accuracy, loss curves) are visualized using Matplotlib and analyzed with Pandas for performance optimization. The chatbot module combines TF-IDF vectorization with intent classification NLP to map user inputs to counselor responses, stored and retrieved via SQLite database. For speech analysis, the system uses Python's Speech Recognition library to process audio inputs and convert them into text. This transcribed text is then transformed using TF-IDF vectorization and passed through a Multinomial Naive Bayes machine learning pipeline for emotion classification. The entire system is deployed as a Flask web application, with k-fold cross-validation ensuring model generalizability across diverse demographic groups. All experimental results, including confusion matrices and precision-recall metrics, are systematically logged in SQLite for continuous improvement.

4.7 MODEL EVALUATION

The system employs model evaluation to ensure reliable performance across all components. For the CNN-based facial emotion detection, key metrics including accuracy, precision, recall, and F1-score are calculated to quantify classification performance, while confusion matrices provide detailed insights into model behaviour across different emotion classes (e.g., happy, sad, angry). The speech emotion recognition module is similarly evaluated using these metrics to validate its ability to correctly identify emotional states from audio inputs. For the chatbot's

NLP pipeline, performance through intent recognition accuracy and response relevance scoring, ensuring alignment with counsellor-approved guidelines. All evaluation results are visualized using Matplotlib/Seaborn for intuitive analysis and stored in SQLite for tracking model improvements over time. Cross-validation techniques prevent overfitting, and real-world testing with diverse user groups further refines the models before deployment.

4.8 MODEL DEPLOYMENT

The system is deployed as an interactive web application built with Python Flask for backend functionality and seamless integration with machine learning models. For real-time analysis, client-side processing enables efficient face and speech emotion detection TensorFlow.js handles in-browser facial expression classification, while Web APIs like the Web Audio API process speech inputs directly on the user's device, ensuring privacy and low latency. The Flask server manages user authentication, data storage in SQLite, and chatbot interactions, delivering a responsive and secure experience. This hybrid approach balances computational load between client and server, optimizing performance for scalable deployment.

SYSTEM TESTING

CHAPTER 5

SYSTEM TESTING

5.1 INTRODUCTION

The goal of testing is to find mistakes. The goal of testing is to find every potential flaw or vulnerability in a work product. It is the process of testing software to make sure it satisfies user expectations and needs and doesn't malfunction in a way that would be unacceptable. Different test kinds exist. Every test type focuses on a certain testing requirement.

5.2 TYPES OF TESTING

1. Unit Testing
2. Functional Testing
3. Integration Testing
4. System Testing
5. End-to-End Testing
6. Security Testing

5.2.1 UNIT TESTING

The facial-powered virtual healthcare system undergoes comprehensive testing to ensure accuracy, reliability, and a seamless user experience. Facial recognition is evaluated under various conditions such as different lighting, angles, and obstructions to ensure precise face detection. Emotion recognition is validated using diverse datasets to accurately identify feelings like joy, sadness, and anger. The depression detection module is tested to assess its ability to recognize prolonged emotional distress and conversational cues that may indicate mental health risks. Voice features, including speech-to-text and text-to-speech, are tested for performance across different accents, background noise levels, and speech clarity. The chatbot is assessed for its ability to hold natural, context-aware conversations and provide meaningful healthcare support. Additionally, user feedback is continuously collected and analyzed to refine system responsiveness and usability. Integration tests ensure all modules work together smoothly, maintaining consistent performance across devices such as smartphones, tablets, and desktops. Security and privacy protocols are rigorously tested to protect sensitive user data, ensuring compliance with healthcare regulations. By simulating real-world scenarios, the system is optimized to respond effectively in both routine and

emergency situations, making it a reliable companion in digital healthcare delivery.

5.2.2 FUNCTIONAL TESTING

Functional testing is done to make sure every part of the facial-powered virtual healthcare system works properly and meets the needs of the project. The facial recognition feature is tested to confirm it can accurately detect and highlight faces, even when lighting or camera angles vary. The emotion recognition module is checked to see if it can correctly read facial expressions and recognize emotions like happiness, sadness, or anger. The depression detection system is tested to ensure it can identify signs of emotional distress and respond appropriately, such as triggering alerts or offering support. The chatbot is also evaluated to make sure it responds naturally and accurately to user questions, keeping conversations clear and helpful. Voice features like speech-to-text and text-to-speech are tested for accuracy across different accents and environments, making sure the system understands and responds well. Overall, this testing ensures the entire system is reliable, easy to use, and able to provide real emotional support and healthcare assistance in everyday situations.

5.2.3 INTEGRATING TESTING

Integration testing is carried out to check how well the different parts of the facial-powered virtual healthcare system work together. For instance, it looks at whether the facial recognition and emotion recognition modules are properly connected making sure that once a face is detected, the emotion recognition module can smoothly analyze the facial data. It also tests the link between emotion detection and the system that identifies signs of depression, ensuring that emotional cues linked to depression trigger the right kind of response or alert. On top of that, the interaction between the voice assistant and chatbot is tested to make sure they stay in sync, understand the context, and give users a smooth, consistent experience as they switch between speaking and chatting. Overall, this testing makes sure all the main components like facial recognition, emotional analysis, depression detection, and voice features work seamlessly together without glitches or miscommunication.

5.2.4 SYSTEM TESTING

System testing is done to make sure the entire facial-powered virtual healthcare system works properly as a complete unit, just like it would in a real-life setting. It checks that all the main features like detecting faces, reading emotions, identifying signs of suicidal thoughts, responding through the voice assistant, and chatting with the user work together smoothly. It also focuses on how users interact with the system, making sure they can easily move from one

feature to another without confusion or technical issues. This kind of testing helps catch problems that might not appear when testing each part on its own. It also checks if the system can handle different situations, like many users at once, emotional variations, or unexpected inputs. The goal is to make sure the system runs reliably, feels natural to use, and gives people the help and support they need in real-world use.

5.2.5 END-TO-END TESTING

End-to-end testing makes sure that the entire facial-powered virtual healthcare system works properly from start to finish. It checks everything, from detecting a person's face and analyzing their emotions to assessing depression and interacting with the voice assistant and chatbot. The goal is to ensure that all the different parts like facial recognition, emotion detection, depression analysis, and the voice assistant work together seamlessly, giving users a smooth experience. This testing also helps catch and fix any potential problems, ensuring the system is reliable and efficient, while making sure sensitive information is kept secure and private.

5.2.6 SECURITY TESTING

Security testing is done to make sure the system keeps sensitive information like personal health details, emotional states, and signs of depression safe and protected. It focuses on checking that data is properly encrypted when it's being sent or received, that facial recognition data is stored securely, and that strong measures are in place to prevent unauthorized access to private emotional or health-related information.

5.3 SUMMARY

Testing is a critical process to identify flaws and ensure a system meets user expectations in real-world conditions. Different types of testing are performed to verify system reliability, integration, and security. Well-designed test cases help uncover issues that could affect the system's performance.

Table 5.1: Test cases for the System

Test Case number	Feature	Input through system	Expected Result	Observed Result
1.	Face Detection	Blurred or noisy facial image	System rejects blurred/noisy image and prompts for a clear capture	Blurred image rejected

2.	Face Detection	Clear facial image	Detect and highlight face accurately	Face detected and highlighted
3.	Emotion Analysis	Clear image with visible facial expressions	System accurately classifies the emotional state from the facial features	Emotional state mentioned correctly
4.	Emotion Analysis	Poor lighting or blurred image	System rejects input and refreshes the page for emotion recognition	Identifies, Rejects, Refreshes the page to capture again
5.	Voice Assistant	User provides a shaky response	System detects stress in voice	Stress detected in voice
6.	Voice Assistant	User provides a calm response	System detects no stress in voice	No stress detected in voice
7.	Chatbot Q&A	User clicks on an emotion block	System displays a list of related questions for the selected emotion	Questions related to the selected emotion block displayed
8.	Chatbot Q&A	User clicks on a question in the selected category	System provides an answer related to the selected question	Answer to the selected question displayed

RESULT AND ANALYSIS

CHAPTER 6

RESULT AND ANALYSIS

The below Figure 6.1 shows the home page. It includes "Login" and "Register" buttons, allowing users to access or create an account to start the analysis. The "Register" button leads to a form where new users can enter their details to create an account. After registering, users can log in using their credentials to access the main interface.

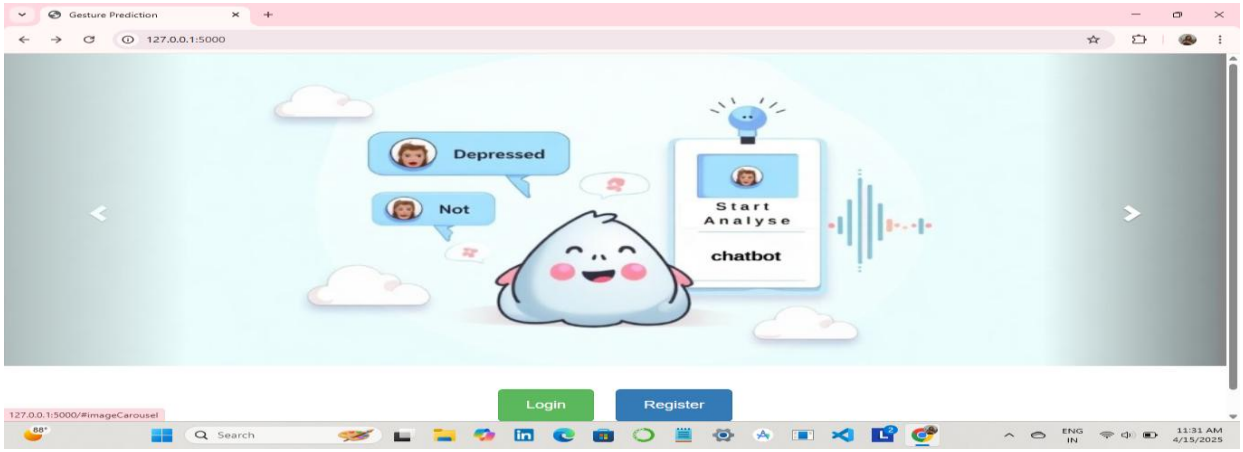

Figure 6.1: Home Page

The below Figure 6.2 shows the dashboard screen, where stress is analyzed through facial expressions and a chatbot interface is available. After a successful login, the user is taken to this screen with a confirmation message. The “Start Analysis” button begins the facial expression capture to assess stress levels. A chatbot button is also available for additional support and guidance.

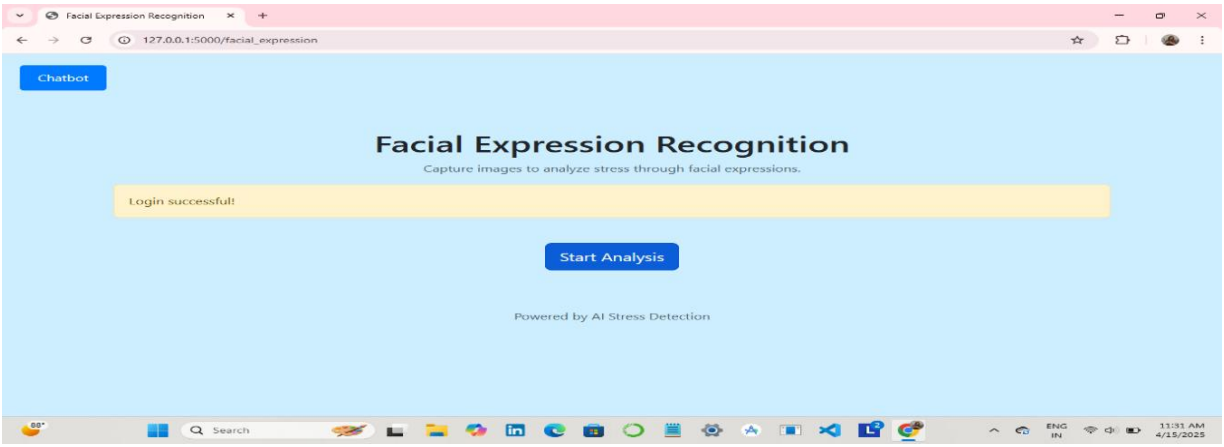

Figure 6.2: Screenshot of Dashboard

The below Figure 6.3 shows the page of facial analysis and speech recognition, which detects the user's facial expressions and analyzes their voice responses. This page presents mental health related questions to assess the user's emotional state. By analyzing both facial expressions and voice responses, the system provides a more accurate evaluation of emotions like happiness, sadness, or stress.

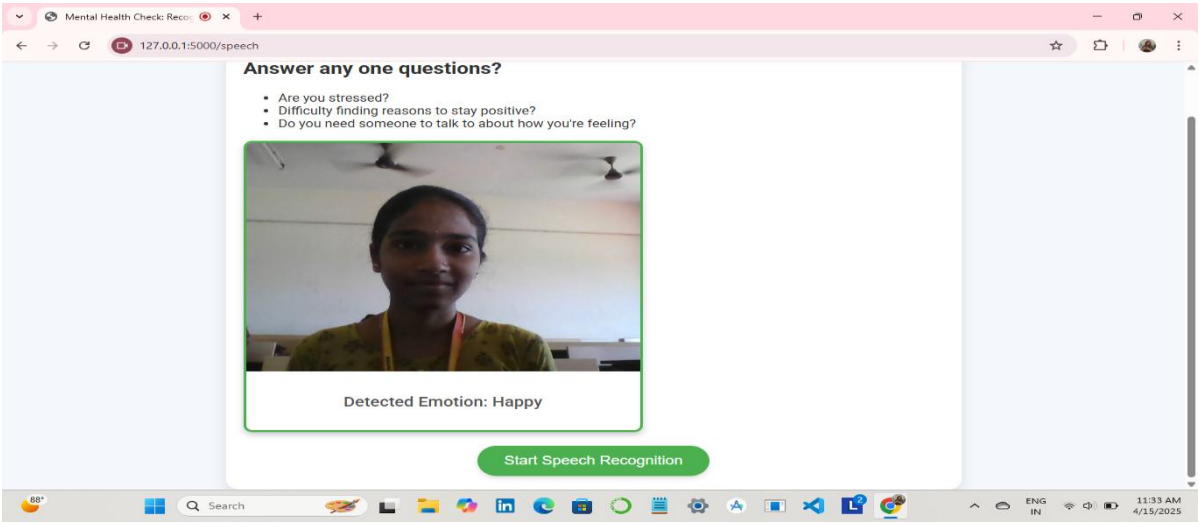

Figure 6.3: Facial Analysis and Speech Recognition

The below Figure 6.4 shows the results of stress evaluation, which are determined using both facial emotion recognition and voice input analysis. The system analysis facial expressions and speech patterns to assess stress levels, with results showing no signs of stress in either modality.

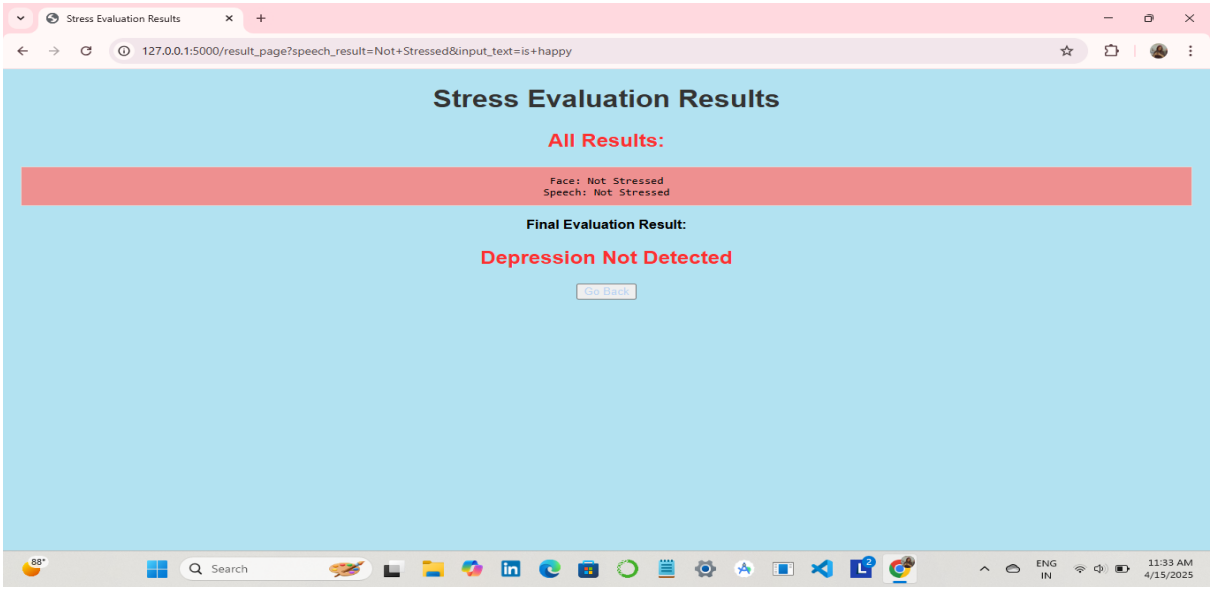

Figure 6.4: Results for Facial Analysis

The below Figure 6.5 shows the interface of the emotion-based Q&A chatbot, where users can select different emotions to interact with the system. The chatbot recognizes multiple emotions like happiness, sadness, and anger, with each option marked by an icon for easy selection.

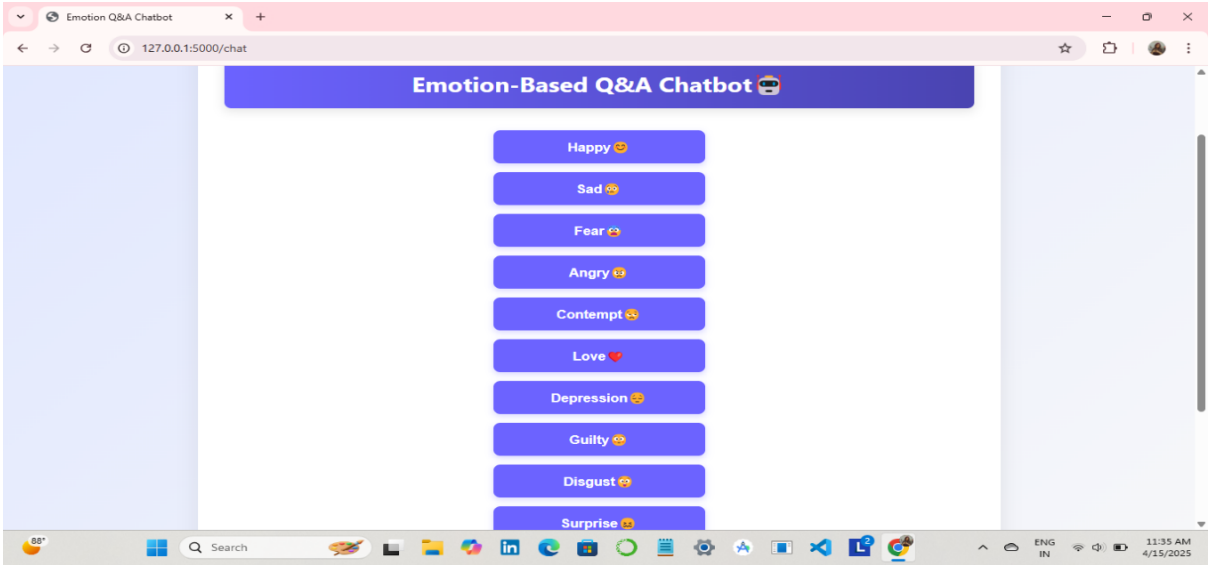

Figure 6.5: Emotion-Based Q&A Chatbot

The below Figure 6.6 shows the chatbot page with predefined Q&As. The image displays a mental health chatbot interface featuring both a list of emotional states like Happy, Sad, Fear, etc and a sample Q&A about depression.

Figure 6.6: Chatbot Predefined Q&A's

CONCLUSION AND SCOPE FOR FUTURE ENHANCEMENT

CHAPTER 7

CONCLUSION AND SCOPE FOR FUTURE ENHANCEMENT

7.1 CONCLUSION

Facial Analysis Powered Virtual Healthcare System presents a transformative approach to digital mental health care by integrating real-time facial emotion recognition, voice analysis, and an interactive chatbot. This multimodal system offers holistic, non-invasive preliminary evaluations of mental well-being, enabling early detection of depressive symptoms and emotional distress. One of its key strengths is delivering instant, personalized recommendations such as guided meditations or referrals to licensed counsellors, empowering both users and clinicians with actionable insights. In addressing the widespread issue of mental health inaccessibility, especially in remote or underserved areas, the system bridges crucial gaps by complementing traditional counselling with intelligent case triage. This enables professionals to prioritize high-need cases efficiently, enhancing both patient care and resource allocation. The platform exemplifies a collaborative synergy between AI and clinical expertise enhancing diagnostic precision without replacing human judgment.

7.2 SCOPE FOR FUTURE ENHANCEMENT

The system holds strong potential for expansion into a comprehensive mental health platform. Future enhancements include AI-powered counsellor matching based on emotional assessments and preferences, enabling seamless virtual therapy sessions. Integrated progress-tracking reports, verified therapist networks, and AI-driven performance analytics will support both user outcomes and professional development. A freemium model with premium visibility tools will help therapists grow their practices while ensuring accessibility. Personalization and user safety remain central: adaptive emotional tracking will refine mood analysis over time, while dynamic therapeutic content like CBT and mindfulness exercises will be tailored to each user. Advanced risk assessment algorithms will detect crisis indicators and trigger immediate alerts for timely intervention, ensuring user safety and clinician responsiveness. These innovations aim to build a robust, scalable, and inclusive mental health ecosystem.

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PERSONAL PROFILE

	Prof.Thapaswini P S She has received her B.E. degree in Computer Science and Engineering from KVG College of Engineering (VTU), Sullia, and her M.Tech. degree in Computer Science and Engineering from VTU Regional Centre, Mysore. She has been actively teaching for the past 8 years. Her areas of interest include Cloud Computing and Data Mining. Email: thapaswinips.cse@gmail.com Phone: 9535622841
	Name: Ms. Sameeksha USN: 4VP21CS078 Address: Samruddhi Hosamane House,Pijakala Kadaba Post,Village and Taluk D.K -574221 Email: sameekshakadaba@gmail.com Phone: 8762294509
	Name: Ms. Swasthika Rai B S USN: 4VP21CS105 Address: Balkady House,Yedamangala Post and Village Kadaba Taluk D.K- 574221 Email: raiswasthika20@gmail.com Phone: 8088123285

	Ms. Thanmayee J USN: 4VP21CS106 Address: “SKANDA”, Basaweshwara Nagara,Near RTO Office, Sirsi, N.K-581402 Email: thanmayeejagadish@gmail.com Phone: 7019023632
	Ms. Pooja A USN: 4VP22CS406 Address: Kuddupadav House,Adyanadka Post,Kepu Village, Bantwal Taluk D.K-574260 Email: poojaasalva@gmail.com Phone: 9901807631

Facial Analysis Powered Virtual Healthcare

Sameeksha[1], Swasthika Rai B S[1], Thanmayee J[1], Pooja A[1], Thapaswini P S[2]
8th Semester, Dept. of Computer Science and Engineering[1]
Assistant Professor, Department of Computer Science and Engineering, VCET, Puttur, Karnataka[2]
Vivekananda College of Engineering and Technology, Puttur, Karnataka, India

***Abstract*— The AI-powered mental health support system aims to revolutionize the mental health care landscape by integrating advanced emotion recognition technologies with real-time virtual assistance. Designed specifically for counselors and mental health professionals, the system employs a multimodal approach combining facial expression recognition, voice analysis, and an intelligent chatbot interface to detect emotional states and possible signs of depression. Using deep learning models, particularly Convolutional Neural Networks (CNNs), facial expressions captured through a camera are analyzed to identify key emotions such as sadness, anger, or anxiety. Simultaneously, a voice assistant evaluates tone, pitch, and speech patterns to detect behavioral cues associated with mental distress. These multimodal emotional insights are then shared with counselors to support accurate and timely intervention. A built-in chatbot further enhances the system by offering empathetic conversations, mental health tips, and regular check-ins, ensuring continuous engagement between therapy sessions. This hybrid model of AI assistance and professional counseling not only improves emotional monitoring and personalized care but also reduces barriers to seeking help. The system ultimately serves as a bridge between traditional therapy and intelligent digital tools, promoting early detection, proactive intervention, and accessible mental health support.**

Keywords*— *Mental health, Deep learning, Convolutional Neural Networks (CNNs), Facial expression recognition, Voice analysis, Emotional monitoring, AI chatbot, Depression detection.

I. INTRODUCTION

Mental health challenges like depression are increasingly prevalent across all age groups, yet traditional assessment methods such as self-reports and clinical interviews often fall short in capturing real-time emotional states. These approaches can be time-consuming, subjective, and inaccessible, especially in remote or underserved regions.

To overcome these limitations, advancements in artificial intelligence offer promising solutions for early mental health screening. Facial recognition and speech analysis technologies enable accurate emotion detection by analyzing expressions and vocal patterns. Emotions such as happiness, sadness, fear, anger, surprise, and disgust can be identified through machine learning models, while vocal cues add further depth to emotional evaluation.

A virtual healthcare platform integrating these technologies allows mental health practitioners to perform quick, automated assessments. The system also features an AI-powered chatbot that engages users in guided conversations, gathering insights into mood patterns and potential depressive symptoms. This enables professionals to prioritize critical cases and deliver timely support.

By combining facial analysis, speech evaluation, and conversational bot, the solution provides a scalable, private, and efficient tool for preliminary mental health assessment, enhancing early detection and promoting accessible care.

II. LITERATURE SURVEY

Sharma et al. [1] proposed a multimodal system combining facial, voice, and text analysis with chatbot interaction for early depression detection. Yu and Kaya [2] enhanced prediction accuracy by focusing on emotionally rich speech segments. Shukla et al. [3] utilized OpenCV and AI for real-time facial emotion recognition. Nirupa et al. [4] developed a CNN-based chatbot that adapts responses to user emotions. Safavi et al. [5] further improved emotion classification by fusing EEG and facial features.

Sharma et al. [6] reviewed machine learning methods for real-time facial emotion detection, highlighting a shift from SVMs to deep learning. Mishra et al. [7] used CNNs like VGG16/VGG19 for high-accuracy facial emotion recognition. Mohan and Chang [8] proposed a lightweight ANN model with attention mechanisms for efficient emotion detection on mobile devices. Yuan et al. [9] employed GRUs to analyze speech and text for detecting depression holistically. Arya et al. [10] compared classifiers (SVM, RF, Naive Bayes) for predicting mental illness, emphasizing scalable early screening.

Yan et al. [11] combined TF-IDF and LexRank to improve text similarity, enhancing sentiment analysis in mental health chatbots. Tripura and Lanka [12] discussed TensorFlow-based architecture for real-time emotion recognition, showing gains in processing speed. Wani et al. [13] created a voice-to-text system using CNNs/RNNs for aiding speech-impaired users. Xu et al. [14] applied deep learning to medical and emotional data for accurate emotional distress detection. Ezerceli and Eskil [15] used CNNs on the FER-2013 dataset for facial emotion recognition, emphasizing model robustness and training.

Ah Ansari Abu Affan Mustafa Kamal, Dhananjay Dinkar Deore, Mubashshera Mirza Gufran Beg, Sufiyan M. Ahsan Ansari, Afeefa Rafeeque, and Md Salman Baig [16] "Facial Emotion Detection and Recommendation using Deep Learning" proposed a system that combines facial emotion detection with personalized video recommendations. The deep learning model used convolutional neural networks (CNNs) to classify emotions and suggest motivational videos based on detected emotions. This innovative approach aims to provide immediate emotional support to users, highlighting the utility of emotion detection in personal well-being.

Kotiyal et al. [17] used pre-trained CNN models like EfficientNetB7 via Keras for emotion detection, demonstrating transfer learning's efficiency for real-time use. Wu et al. [18] developed an end-to-end deep network for real-time face and landmark detection, crucial for responsive emotion recognition. Suryamritha et al. [19] employed a CNN-LSTM model on the RAVDESS dataset to enhance speech-based emotion and speaker recognition. Gurunath Shivakumar et al. [20] improved ASR systems by integrating large speech-text models, boosting transcription accuracy for emotion-aware applications like mental health chatbots.

Chandel et al. [21] built a healthcare chatbot leveraging AI and NLP for personalized symptom analysis and mental health support. Dhanasekar et al. [22] created an emotion-aware chatbot for student mental health, especially useful during the pandemic. Zhou et al. [23] enhanced speech synthesis with mixed emotions to improve natural interactions in virtual healthcare. Fatima et al. [24] used a CNN-BiLSTM model to detect depression from social media, enabling early intervention. Abinaya et al. [25] designed an emotion-aware conversational agent that adapts responses based on user behavior for empathetic mental health support.

III. PROPOSED SYSTEM

The system offers users two modes of interaction: real-time facial emotion recognition with voice input and a chatbot-driven conversation based on user-selected emotions. By integrating multimodal data sources, it enhances the accuracy of emotional assessment and ensures a more comprehensive understanding of the user's mental state. The facial emotion recognition module can classify expressions into categories such as happy, sad, angry, or neutral, enabling early identification of potential depressive symptoms. Meanwhile, the chatbot serves as a supportive tool, guiding users through predefined questions and responses tailored to their emotional condition. This combination of AI-driven analysis and conversational support makes the platform a valuable tool for remote mental health monitoring, especially for underserved populations.

IV. METHODOLOGY

The system also features a user-friendly interface that guides individuals through each step of the interaction, ensuring ease of use even for those unfamiliar with digital platforms. Real-time emotion recognition is achieved through seamless integration of webcam and microphone inputs, providing instant feedback based on the user's expressions and tone. The chatbot is designed to simulate human-like conversations, promoting emotional comfort and encouraging users to express their feelings openly.

Additionally, the system includes built-in privacy measures, ensuring that sensitive data such as emotional assessments and user conversations are securely stored and handled. Through this simulation, the effectiveness of combining AI-powered analysis with compassionate

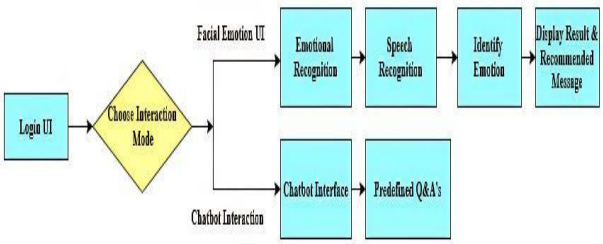

Figure 1: Architecture of the System

Figure 1 is about system architecture which includes a user interface, emotion detection module, chatbot, and backend system for smooth data flow and session management.

In the real-time mode, the system utilizes a webcam and microphone to capture facial expressions and voice inputs, which are processed using Convolutional Neural Networks (CNN) and natural language processing (NLP) techniques to detect

emotional cues such as stress, sadness, or happiness. The analysed emotion data is then used to trigger tailored feedback such as relaxation tips, motivational messages, or prompts for further support.

The flowchart shown in Figure 2 represents the operational flow of the intelligent emotion-aware system, providing a structured overview of how the application functions from user entry to emotional response generation. The process initiates with a secure registration or login step, ensuring that each user's session and data are uniquely tracked and stored. Once authenticated, users are directed to select one of the two primary modes of interaction: real-time emotion recognition or chatbot-based communication.

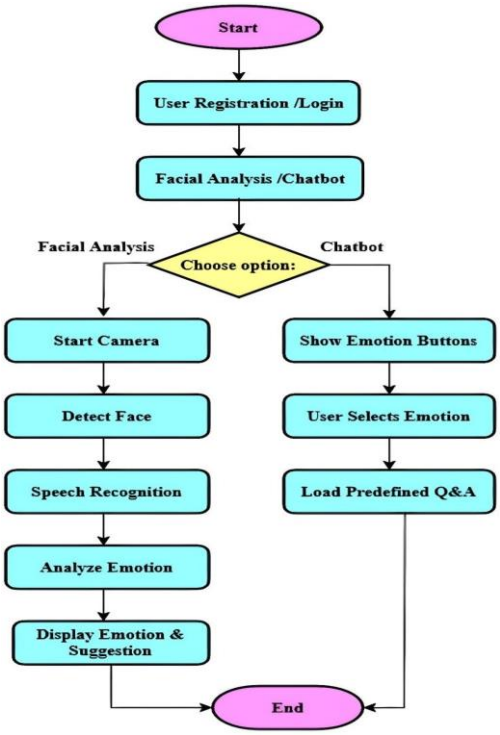

Figure 2: Flowchart of the Project

To start, the data collection process involves leveraging publicly available benchmark datasets such as FER-2013 for facial emotion recognition and tweet_emotion.csv for speech sentiment analysis, which enable accurate model training. User-generated inputs, including facial expressions, voice samples, and chatbot interactions, are also securely stored to support future retraining and continuous improvement. The system preprocesses data through a multi-step workflow images are resized, normalized, and augmented for facial detection, while text and speech inputs are transcribed and vectorized using TF-IDF for sentiment analysis. Facial emotions are classified using a Convolutional Neural Network (CNN) with layers such as Conv2D, ReLU, MaxPooling, Dropout, Flatten, and SoftMax, while speech emotion detection relies on Python's Speech Recognition library and a Multinomial Naive Bayes classifier. The Haar Cascade algorithm aids in face detection, and chatbot responses are matched via cosine similarity to ensure relevant support. Model training and validation are performed using Python, TensorFlow, Keras, OpenCV, and scikit-learn, with performance monitored through metrics like accuracy, precision, recall, F1-score, and confusion matrices, and visualized using Matplotlib and Seaborn. Evaluation also includes k-fold cross-validation and testing with diverse user groups to ensure reliability. Finally, the system is deployed as a responsive web application using Flask for backend services, TensorFlow.js for client-side facial expression analysis, and Web APIs for real-time speech processing—delivering an efficient, secure, and scalable virtual healthcare platform.

V. RESULTS

The developed virtual healthcare system effectively integrates facial emotion recognition, voice-based sentiment analysis, and a chatbot interface to assess and support users' mental health. Starting with a secure registration and login process, users can undergo emotion evaluation through facial expressions and voice responses to specific mental health-related questions. The system analyses this multi-modal input to determine signs of stress or depression, offering accurate emotional assessments. In addition, the platform features an emotion-based chatbot that enables users to engage with predefined Q&A tailored to different emotional states such as happiness, sadness, and fear.

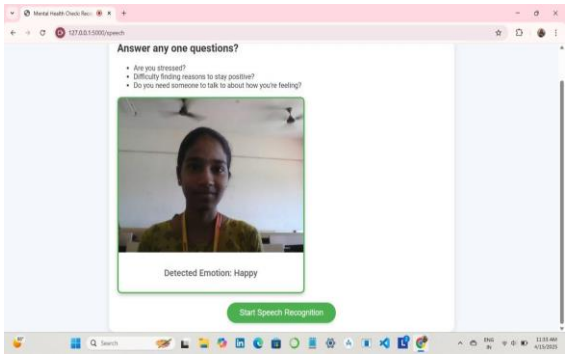

Figure 3: Facial and emotion Analysis

In Figure 3 Both facial and vocal inputs are combined to increase the reliability of emotion detection, offering a more holistic view of the user's psychological state. The system then delivers personalized suggestions, such as prompts for mindfulness or relaxation, based on the analysis.

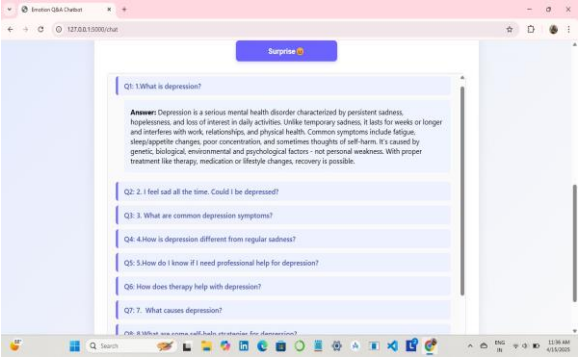

Figure 4: Chatbot with Pre-defined Q&A

The Figure 4 is about the user engagement by offering emotion-specific questions and responses, creating a more empathetic and supportive environment. The seamless navigation from login to analysis and interaction ensures a smooth user experience, while the results provide clear insights into the user's emotional state. Overall, the platform demonstrates how artificial intelligence can be harnessed to deliver accessible, responsive, and intelligent mental health support in a digital setting.

VI. CONCLUSION AND FUTURE WORKS

Facial Analysis Powered Virtual Healthcare System offers an innovative approach to mental health assessment by combining real-time facial emotion recognition, voice analysis, and chatbot interaction. This multimodal solution enables early detection of depressive symptoms, delivering personalized suggestions like guided meditation or counselling referrals. Designed to bridge mental health gaps, especially in remote areas, the system supports counsellors by streamlining triage, ensuring high-need cases are prioritized. It enhances diagnostic precision while maintaining human clinical judgment, promoting better patient outcomes and resource management.

Looking ahead, the system aims to evolve into a comprehensive mental health platform. Future plans include AI-powered counsellor matching, automated progress reports, and verified therapist networks to expand professional reach. Personalization will be deepened through adaptive emotional tracking and customized therapy content such as CBT exercises and mindfulness practices. Safety features will include real-time risk detection and alerts for crisis intervention, ensuring timely support. These enhancements will create a scalable, user-centered ecosystem that benefits both clients and mental health professionals.

Acknowledgment

We are pleased to submit this project paper on the Facial Analysis Powered Virtual Healthcare System. We would like to express our sincere gratitude to our project guide, Professor Thapaswini P S, for her invaluable guidance and continuous support throughout the project.

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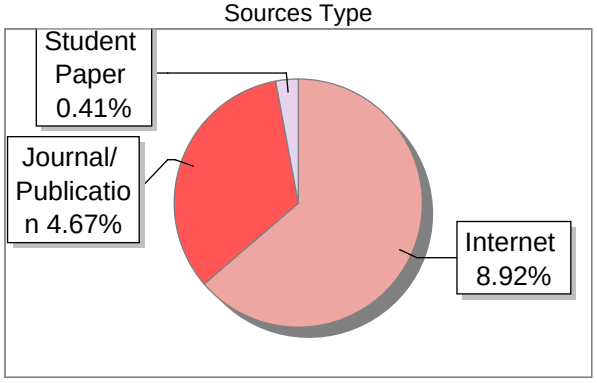

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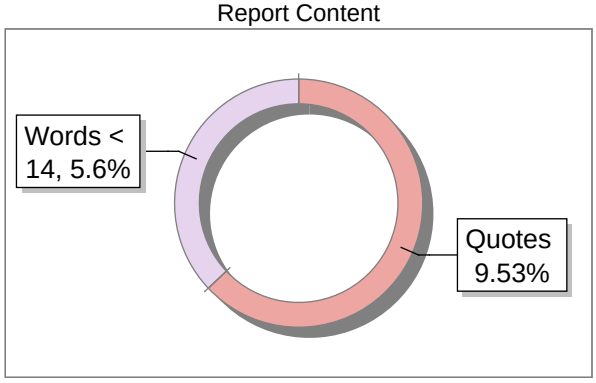

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